

Lights Out Trivia

Project Summary

Our project is a Formula 1 trivia website that generates questions using real F1 data instead of relying on a fixed list of questions. A problem with most trivia websites is that after playing them a few times, users start seeing the same questions again. We want to avoid that by generating questions from data about races, drivers, constructors, standings, qualifying, lap times, and other F1 statistics.

Users will be able to play generated quizzes, choose topics such as a specific driver, team, or era, create their own quizzes, save questions they like, and compete on leaderboards. Since the questions come from the data in our database, we can also update the application as new races happen instead of having to manually rewrite questions and answers whenever a record changes.

Creative Component

The main creative component of our application is the dynamic question generation system. Instead of storing a fixed set of trivia questions, the backend will generate questions by running queries on the F1 datasets. For example, it could find race winners at specific circuits, compare driver results across seasons or generate questions based on standings and lap times.

The system will also generate the correct answer and possible wrong answers directly from the data. We also plan to use a user's previous quiz history to reduce repeated questions and adjust the difficulty over time. This will require combining data from multiple tables and turning the query results into questions that make sense to the user.

Usefulness

- Why our application is useful:
 - The Problem:
 - Even though there are tons of F1 trivia websites out there, hand-written trivia always runs out. Whenever a quiz is played once, it loses all its meaning since the user now memorizes the questions and its replay value dies. This hurts user retention and also puts a lot of pressure on the site owner who has to keep making new quizzes every week.
 - The answers for questions keep changing as records break. For example, the answer for the question "Which driver has won the most number of

“races all time?” might be driver A today but can become driver B tomorrow once they surpass them.

- Our Solution:
 - With the help of dynamic question generation and real datasets containing accurate historical data, our application makes sure that questions are generated in real-time rather than just creating a set of X questions and asking them to the user over and over again. This means that every new quiz poses a new challenge and the user can replay it an infinite number of times without getting bored due to repeated questions.
 - As new races happen and race data comes in, we keep updating the dataset to include the latest data and times. This solves the problem of outdated answers since our application is pulling the latest directly from the dataset rather than relying on hardcoded answers.
- Basic Functionality:
 - Users can:
 - Sign up and create their accounts
 - Browse and play system generated quizzes
 - View global and location-wise leaderboards
 - Build custom quizzes and share them with friends
 - Compete against friends 1 on 1
 - Simple Features:
 - Account management
 - Browsing quizzes
 - Maintaining leaderboards
 - Complex Features:
 - Dynamic question generation from the database
 - Filtering questions based on the topic selected by the user
 - Choosing the questions for the daily challenge based on the user's difficulty level
- Similar applications/websites:
 - Sporcle F1 Quizzes: Community made quizzes on Formula One on various topics (e.g. circuits, race winners, etc.)
 - Limitations: Hardcoded questions, repeating a quiz produces the same questions, no replay value
 - Solution: Dynamically generated questions from the dataset, removing repetition and the user can replay the quiz however many times they want without getting bored
 - Official F1 App Occasional Quizzes: A few questions posted once a while in the app to create a gamified experience and keep the user engaged
 - Limitations: Only a limited amount of questions available in the question bank, quizzes are only released once in a while, the user can't select the topics/eras they want to play trivia on

- **Solution:** The user can select any era/topic they want to test their knowledge on and the app never runs out of questions since they are automatically generated and don't require any manual effort.

Realness - Data Sources

Primary Dataset Features:

1. **Source:** The data is hosted on Kaggle as the [Formula 1 World Championship \(1950 - 2024\)](#) dataset, originally compiled from the Ergast Motor Racing Developer API.
2. **Format:** The data is provided in **CSV** format, structured across a relational database of 14 separate files.
3. **Data Size (Cardinality & Degree):** The overall dataset is approximately 22 MB and features a combined degree of **120** columns across all tables. The primary table has a degree of 18 columns and a cardinality of over **26,000** rows.
4. **Information Captured:** The dataset captures comprehensive historical racing data spanning from the inaugural 1950 season through 2024. This includes highly detailed metrics on race results, driver and constructor standings, qualifying times, circuit details, lap times, and pit stops.

Secondary Dataset Features:

1. **Source:** The data is hosted as an API ([FastF1](#)), which is scrapable and a slice of the dataset can be found on Kaggle: [Formula 1: Race Data and Telemetry \(Updatable\)](#)
2. **Format:** API usage. We will scrape the relevant data to load it in our database.
3. **Granularity and Row Count:** Telemetry data is sampled 10 times per second. With the average race time around 1.5 hrs and avg drivers count on the grid as 20 drivers, there ~1M rows per race session and we are looking over the entire F1 history when telemetry was available(post 2018) till date so we would have more than **1B** records for telemetry data. As part of the course project we would sample from this data (instead of 10 times per second, we can pull data every 2 seconds reducing the dataset by 20x). While the dataset does not post any official degree, we can see about ~**100** columns across the various data objects in the API.
4. **Information Captured:** The dataset captures high-resolution racing telemetry and lap-based aggregates. This includes detailed metrics such as lap times, sector splits (Sector 1, Sector 2, and Sector 3 times), tire compound usage, tire life, and broader race metadata regarding track conditions.

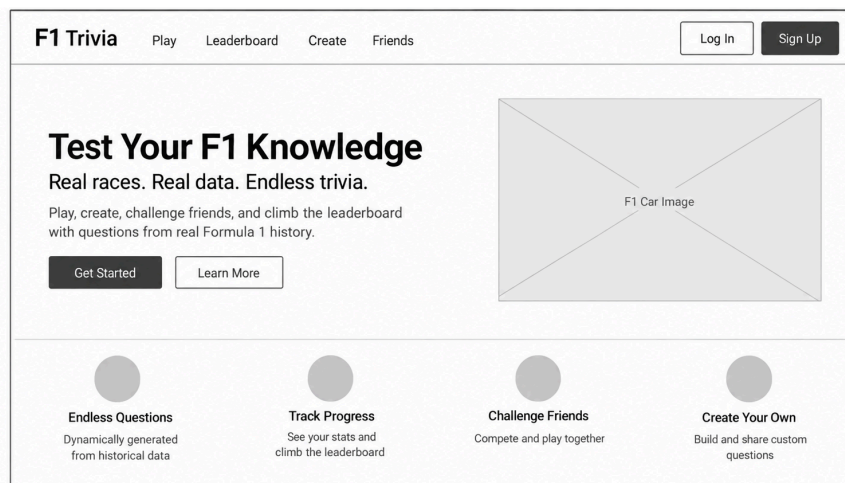
Functionality Description

1. **User Focused**
 - a. When the user launches a trivia round, the backend executes advanced queries on the datasets to come up with dynamic questions.

- b. The user can choose filters like team, driver, era to generate a focused quiz on the topic.
 - c. The user can create a new quiz with pre-loaded questions or new questions that they come up with.
 - d. The user can delete a previously created quiz
 - e. The user can update a previously created quiz
 - f. The user can [create/update/read/delete] a favorite questions list once they see a question to review later or use it in a custom quiz
 - g. The user can share the quiz with their friends to challenge them
 - h. The user can compete for the top spot at the leaderboard
 - i. A daily challenge that creates 5 questions specific to the user (difficulty matched) which the user has not seen/attempted in the past X days.
 - j. The user can search for quizzes by title, driver, team, era, or category.
- 2. Admin**
- a. The admin can create a record of the latest race data
 - b. The admin can update any past historical data in case it is erroneous

Low-Fidelity UI Mockup

1) Landing Page



2) Login

F1 Trivia

PlayLeaderboardCreateFriends

Log InSign Up

Sign Up

Log In

Username

Email

Password

Create Account

or

G

Continue with Google

Already have an account? [Log In](#)

F1 themed image/graphic

3) QUIZ UI

F1 Trivia

PlayLeaderboardCreateFriends

User

Question 3 of 10

Time: 00:15

Which driver won the 2010 Canadian Grand Prix?

☐ Lewis Hamilton

☐ Jenson Button

☐ Sebastian Vettel

☐ Fernando Alonso

Submit Answer

F1 car image

4) Result and Progress tab

F1 Trivia

PlayLeaderboardCreateFriends

User

Correct!

Jenson Button won the 2010 Canadian Grand Prix at Circuit Gilles Villeneuve.

Next Question

View More Info

Fun Fact:

This was Jenson Button's first win with McLaren.

Track image

Your Progress

Questions Answered

3 / 10

Correct Answers

3

Current Streak

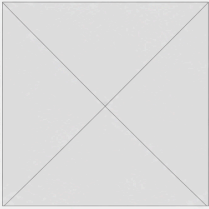
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5) Leaderboard UI

F1 Trivia Play **Leaderboard** Create Friends User ▾

Global Leaderboard Global Friends

#	Username	Points	Correct %	Games Played
1	MaxVerstappenFan	12,450	92%	320
2	F1Nerd	11,980	89%	298
3	PitStopPro	11,230	87%	275
4	FerrariFaithful	10,940	85%	260
5	GridGuru	10,521	84%	241
6	You	9,870	83%	210
7	RaceDayReady	9,320	82%	205
8	CheckeredChris	8,990	80%	190
9	TurboTrivia	8,410	78%	176
10	ApexHunter	8,120	77%	168



Podium / F1 image

Your Stats

Rank	#6
Points	9,870
Correct %	83%
Games Played	210

Play Again

6) Create quiz UI

F1 Trivia Play Leaderboard **Create** Friends User ▾

Create a New Quiz
Build your own F1 trivia question using real historical data.

Question
Enter your question here... 0/200

Answer Options

A	Enter option A...	☐ Correct Answer
B	Enter option B...	☐ Correct Answer
C	Enter option C...	☐ Correct Answer
D	Enter option D...	☐ Correct Answer

Explanation (Optional)
Add a short explanation or fun fact... 0/300

Tags (Optional)
e.g. 2021, Hamilton, Monza
Add keywords to help categorize your question

Save Question

Cancel

Preview
Which driver won the 2010 Canadian Grand Prix?

A	Lewis Hamilton
B	Jenson Button
C	Sebastian Vettel
D	Fernando Alonso

Additional Settings

Category General ▾	Difficulty Medium ▾
☐ Make this question public (visible to other users)	
☐ Allow friends to use this question	

Project Work

- **Krishna**
 - Responsible for the dynamic question generation system.
 - Will build backend logic that queries the F1 datasets and converts the results into trivia questions.
 - Will also work on quiz creation, updating, deleting, and search functionality.
- **Chirag**
 - Responsible for the database design and data loading.
 - Will set up the main database tables and relationships.

- Will work on importing and cleaning data from the Kaggle dataset and FastF1.
- Will also work on backend queries involving drivers, races, constructors, and telemetry data.
- **Dhruv**
 - Responsible for quiz attempts, results, and progress tracking backend.
 - Will build backend functionality for recording user answers, calculating quiz scores, and storing quiz history.
 - Will also work on the frontend for the landing page, login, quiz interface, and results pages.
 - Will connect these pages to the backend APIs.
- **Rohan**
 - Responsible for user accounts, leaderboards, and user-specific features.
 - Will work on login and account-related backend functionality.
 - Will build backend functionality for leaderboards, favorite questions, and daily challenges.
 - Will also work on admin functionality for adding and updating race data.

The backend work will be distributed across all four team members. Krishna will focus on quiz generation and quiz CRUD operations. Chirag will focus on the database, data ingestion, and dataset queries. Dhruv will focus on quiz attempts, scoring, and quiz history. Rohan will focus on user accounts, leaderboards, favorites, daily challenges, and admin functionality. Frontend work will be shared as needed while each member remains responsible for their assigned backend features.